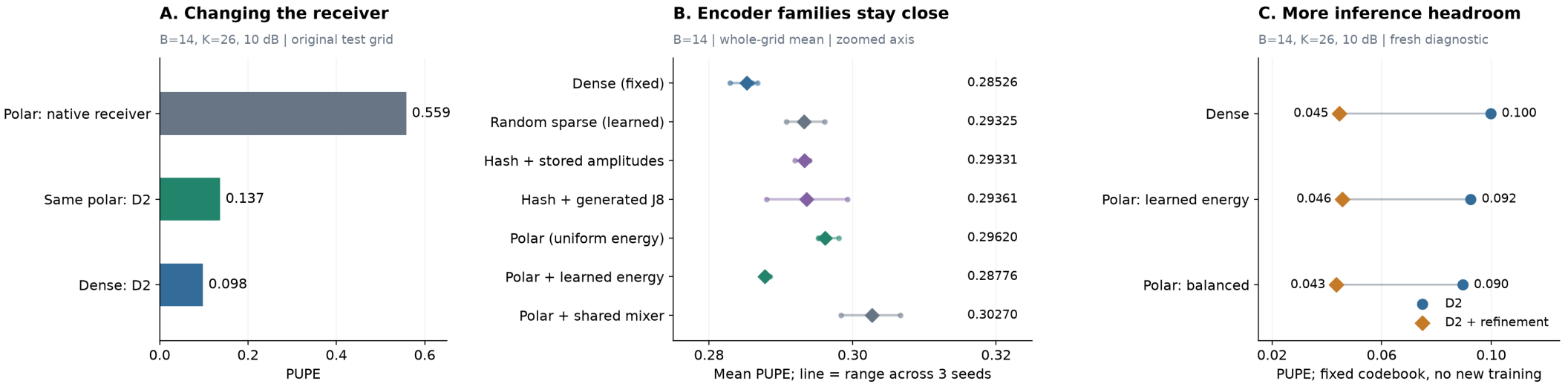
Why we now focus on polar multiuser decoding

The largest measured gain came from better inference. A modest energy correction then brought polar close to dense.



Real AWGN, n=256, known K/noise. Native scheme is a small-B adaptation, not a paper-scale reproduction.
A/B: 3 seeds; 256 frames/cell/seed. B: K=7,15,22,26; -4:2:10 dB; all use trained D2. Only dense/uniform polar have fixed encoders.
C: 512 fresh frames/seed; balanced polar uses transferred D2. D2 and likelihood refinement still scan the full alphabet.

Pause the hash search

Hash J8 retains nearly the same mean PUPE as its stored-amplitude parent: **0.29361 vs 0.29331**, with 64x fewer amplitude values at B14. It still needs a multiuser inverse. This is a priority decision: compact encoding alone did not solve recovery.

Keep polar; simplify energy shaping

Learned polar energies reach **0.28776 vs dense 0.28526** over the full B14 grid, and improve every seed at B12/B14. Balancing explains much of the gain. The shared mixer worsened these runs. **Placements stayed fixed**; learning or removing ODMA remains untested.

Spend the next effort on inference

On identical polar waveforms, D2 cuts stress-point PUPE **0.559 to 0.137**. A later observation-only refinement roughly halves D2's remaining error for dense and shaped polar. The current work derives a joint polar receiver with explicit accuracy and cost accounting.

Scope: historical small-B adaptations; real AWGN; unit-energy waveforms; known K/noise. D2 and refinement enumerate the alphabet. Close means do not prove equivalence or optimality. Polar gives useful structure; scalable multiuser decoding remains the research problem.

Sources: Jobs 032-034; original grids plus 10 Oct paired diagnostic. Plot data and source hashes saved alongside. | 18 October 2026